

SOFTWARE ARCHITECTURES INTRODUCTION

Software Engineering Seminar

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Outline

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1 Systems Thinking

2 Reference Architectures



Outline

1 Systems Thinking

2 Reference Architectures



What is a System?

- A **system** is a set of interacting components that work together to achieve a common goal.
 ↳ *modules - classes*
- A **system** is a collection of elements that are organized in a specific way.
 ⇒ *architecture*
- A **system** is a structure that is designed to perform a specific function.

A-B-C

.

*A-B
O*

*B
A-C*



Systems Analysis Process

- **Systems analysis** is the *process* of **studying** a **system** in order to identify its **components**, **interactions**, and **goals**.
- **Systems analysis** is the *process* of **understanding** how a **system works** and how it can be **improved**.
- **Systems analysis** is the *first step* in the **systems development lifecycle**.

Car



LLM
fire turning



Systems Analysis Techniques

- **Systems analysis** uses a variety of techniques to study a system.
- It includes interviews, surveys, observations, and document analysis.
- It also includes data modeling, process modeling, and requirements analysis.

final users

stakeholders

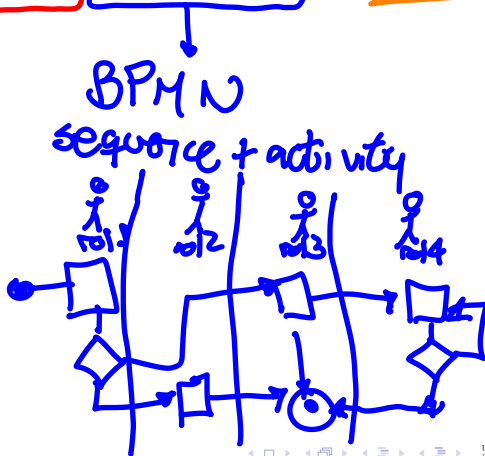
business processes

flowchart

diagrams



- **Systems analysis** uses a variety of techniques to study a system.
- It includes interviews, surveys, observations, and document analysis.
- It also includes data modeling, process modeling, and requirements analysis.



- Sources
- Format + structure
- Understanding



Complexity and Emergence

- **Complexity** is the *degree* to which a **system** is difficult to understand.

- Emergence is the *appearance of unexpected properties* in a system

- They form from interactions of its components
- Complexity and emergence are common in dynamic systems that are non-linear.

- They can be studied and understood through systems analysis and modeling.

+ elements
+ relations

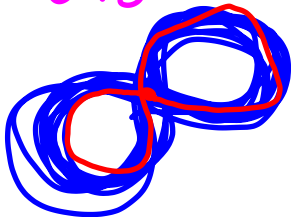
chaos
theory

randomness
(stochastic)

non-linear

chaotic
attractors

Lorenz



Error runtime

- **Complexity** is the degree to which a system is difficult to understand.
- **Emergence** is the appearance of unexpected properties in a system that arise from the interactions of its components.

Swarm intelligence



Complexity and Emergence

- **Complexity** is the *degree* to which a **system** is **difficult** to **understand**.
- **Emergence** is the *appearance* of **unexpected** properties in a **system** that **arise** from the **interactions** of its **components**.
- **Complexity** and **emergence** are common in dynamic systems that are non-linear and chaotic.
- They can be studied and understood through systems analysis and modeling.

Complexity → elements + relations
 Emergence → rules + randomness



Strategies to Solve Problems

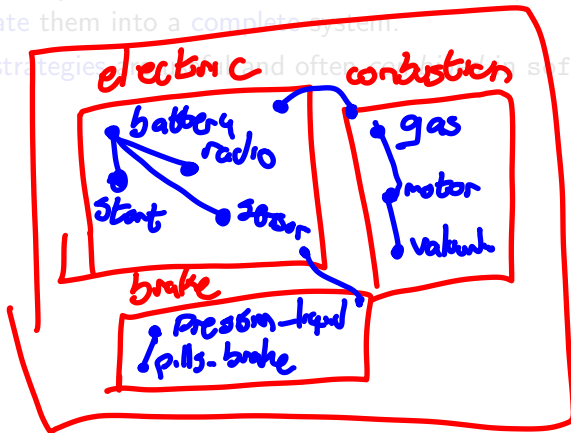
- **Top-Down:** Start from the big picture and break it down into smaller parts.

G.O.P.

- **Bottom-Up:** Start from small, well-defined components and integrate them into a complete system.

Car

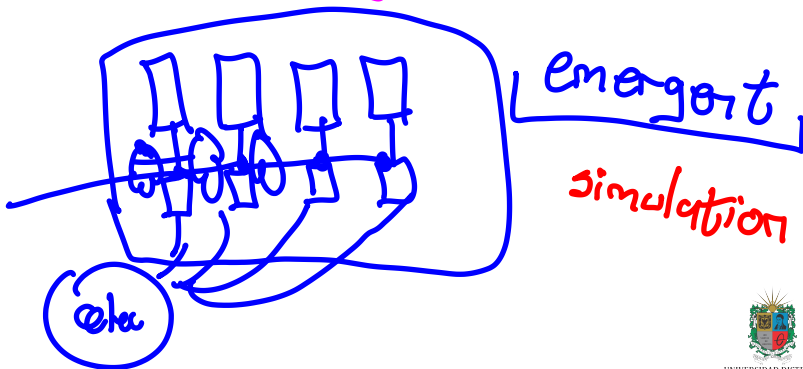
- Both strategies are used and often combined in software design.



Strategies to Solve Problems

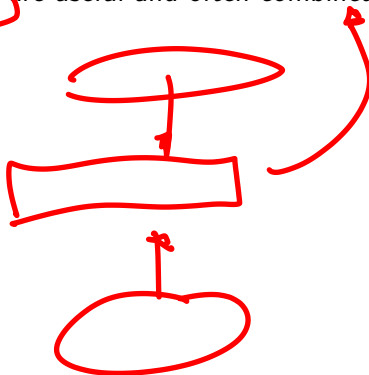
- **Top-Down:** Start from the **big** picture and **break** it down into smaller parts.
- **Bottom-Up:** Start from **small**, well-defined components and integrate them into a **complete** system.

↳ more understanding / more difficult



Strategies to Solve Problems

- **Top-Down:** Start from the **big** picture and **break** it down into smaller parts. *→ Software*
- **Bottom-Up:** Start from **small**, well-defined components and **integrate** them into a **complete** system.
- **Both strategies** are useful and often *combined* in software design.



Process Definition

- A **Process** is a **series** of steps or actions taken to achieve a particular end.
- **Processes** are used to **organize** and **manage** work.

algorithm → code
business/
no tech



software
solution



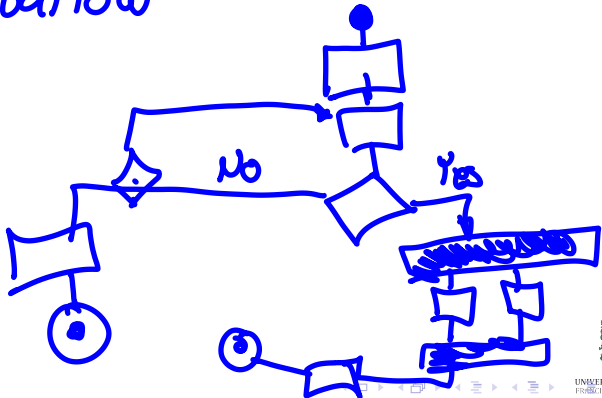
Workflows

business processes

- A **Workflow** is a series of tasks that are performed in a specific order to *achieve a goal*.
- **Workflows** are used to automate and optimize business processes.
- **Workflows** can be sequential, parallel, conditional, or repetitive.

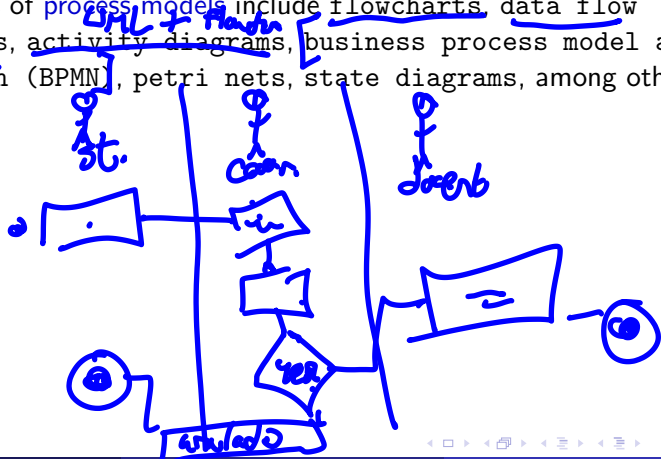
steps | code

ETL - DataFlow



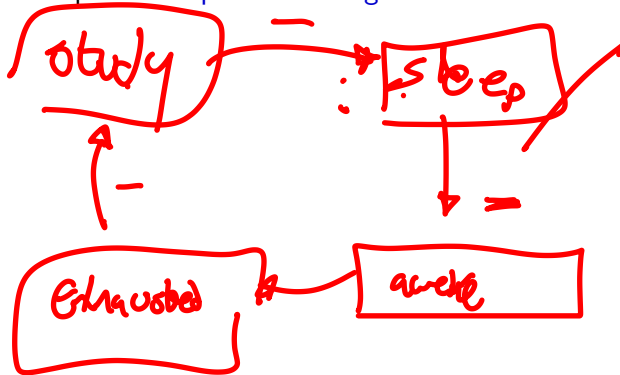
Process Models

- A **Process Model** is a representation of a **process** that shows the sequence of steps and the **relationships** between them.
- **Process models** are used to **analyze**, **design**, and **improve** processes.
- Examples of **process models** include **flowcharts**, **data flow diagrams**, **activity diagrams**, **business process model and notation (BPMN)**, **petri nets**, **state diagrams**, among others.

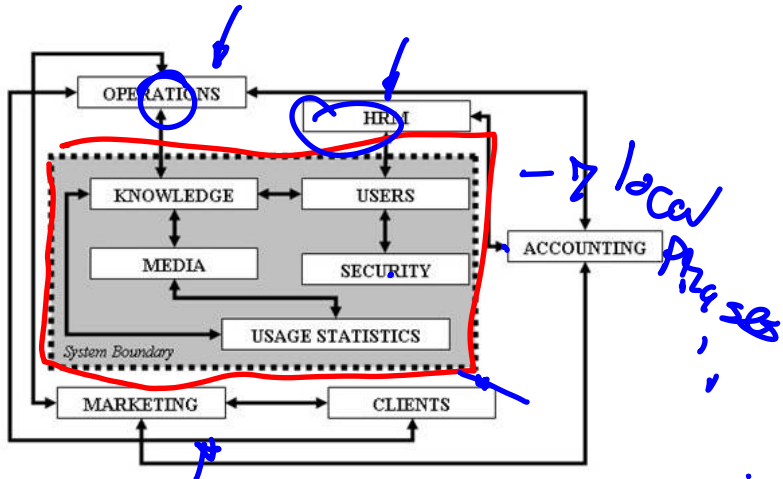


Causal Loops

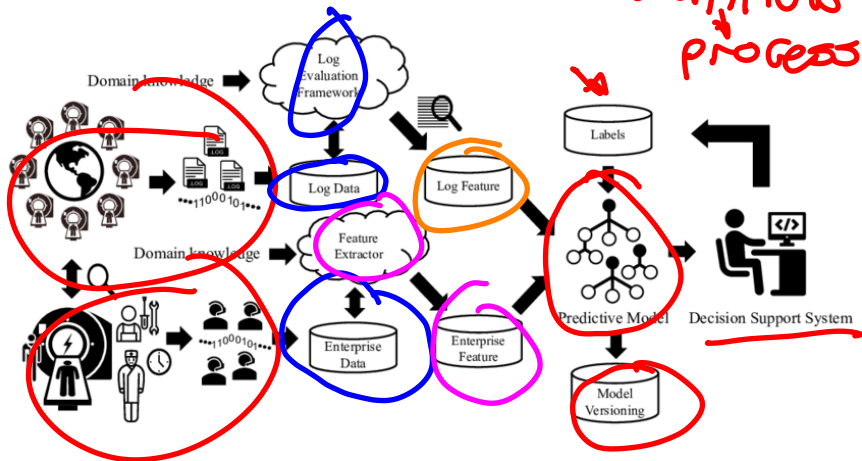
- A **Causal Loop** is a diagram that shows the relationships between different variables in a system.
- Causal loops are used to analyze and understand the dynamics of a system.
- Causal loops can be positive or negative.



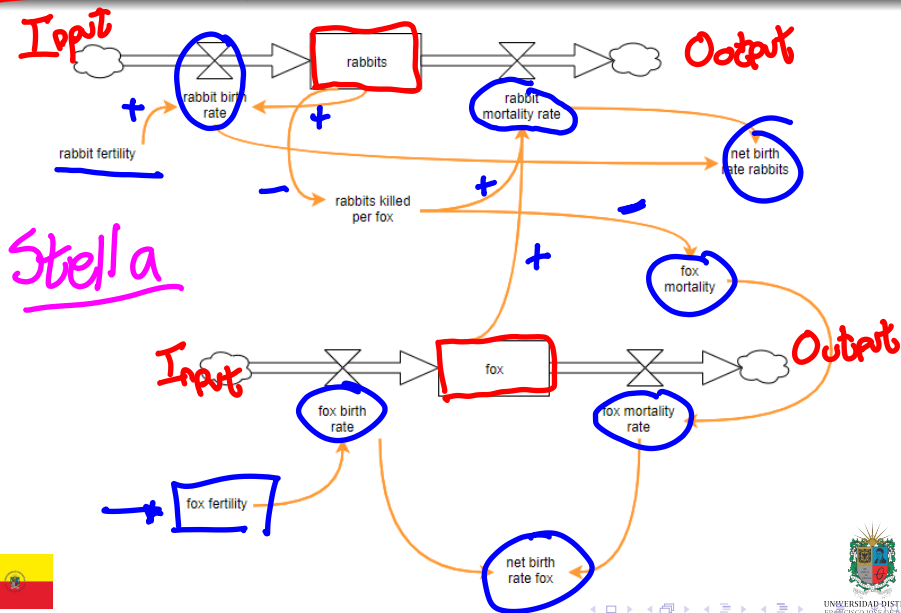
System Schema Example: Company Structure



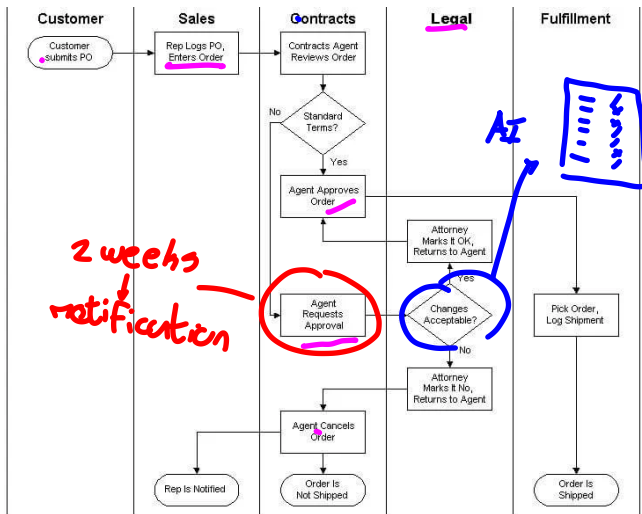
System Schema Example: Processing Pipeline



Stock and Flow Diagram



Business Process Model and Notation (BPMN)



Outline

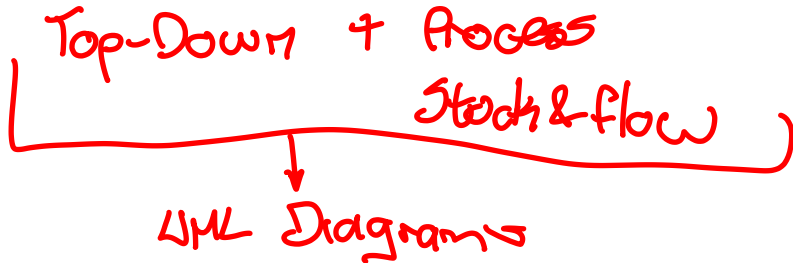
1 Systems Thinking

2 Reference Architectures



Design Before Code

- **Design** should come *before coding*.
- Jumping into *code without a plan* leads to *confusion* and *rework*.
- **Good design** *clarifies* the problem and *guides* the solution.



Understanding the Requirements

- **Requirements** must be well understood before design.
- Ask questions, clarify ambiguities and document all requirements.
- Requirements define the scope and direction of the design.

Implications - Trade-offs



Design Based on the Problem

- **Design** should be driven by the problem, not by technology.
- Focus on what **needs** to be solved, *not just* how to implement it.
- Use the **problem statement** to identify **key objects** and their *relationships*.

business

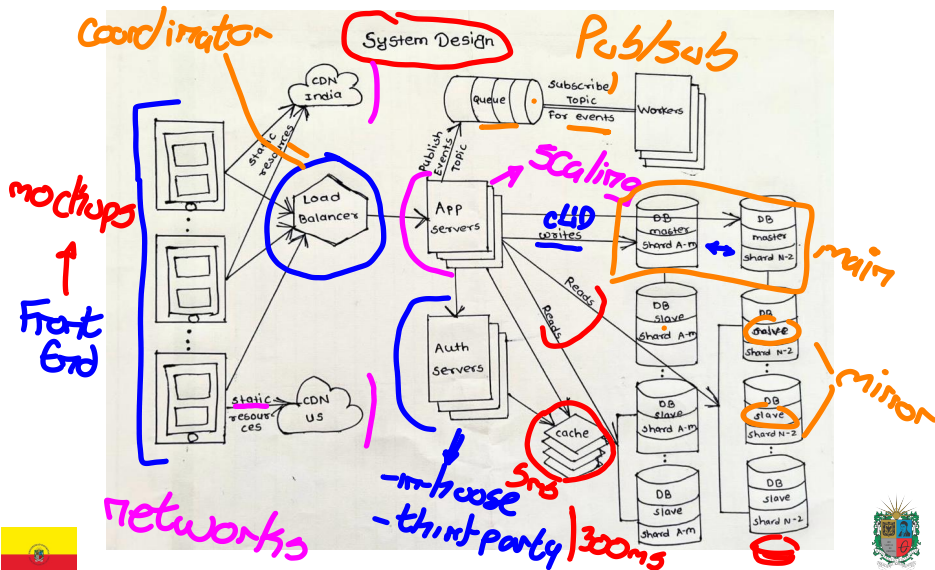


system

identify
problems



Systems Design applied to Software Architectures



Conceptual Design and Technical Design

- **Conceptual Design:** What the system should do using high-level models.
- **Technical Design:** How the system will be implemented, using detailed diagrams and specifications.
- Both are essential for a successful software project.

understand
goals
+
scope

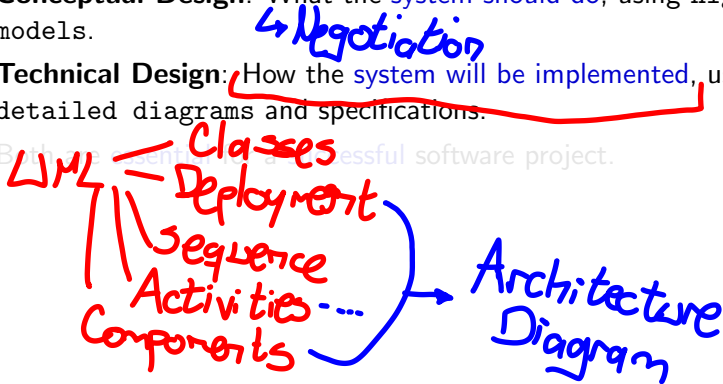
all stakeholders

- Mockups
- CRC cards
- Business Process
BPMN



Conceptual Design and Technical Design

- **Conceptual Design:** What the system should do, using high-level models.
- **Technical Design:** How the system will be implemented, using detailed diagrams and specifications.
- Both are essential for a successful software project.



- **DataBase System Design**



Conceptual Design and Technical Design

- **Conceptual Design:** What the **system should do**, using high-level models.
- **Technical Design:** How the **system will be implemented**, using detailed diagrams and specifications.
- Both are **essential** for a **successful** software project.

Scope
Implications

tech team
clients
all in the same page



What is a System Architecture?

- A **system architecture** is the structure of a system that defines its components, interactions, and relationships.
- A **system architecture** is the blueprint of a system that guides its development and implementation.
- A **system architecture** is the foundation of a system that ensures that it meets the needs of its users.

Solve a Problem



Types of System Architectures

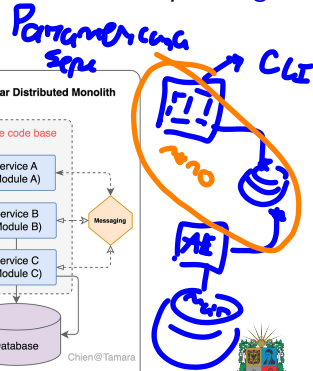
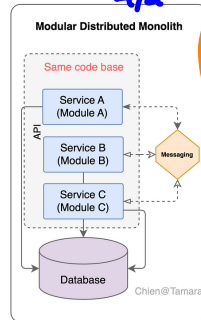
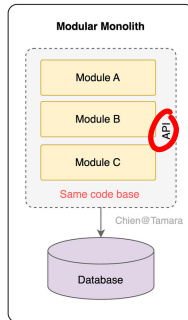
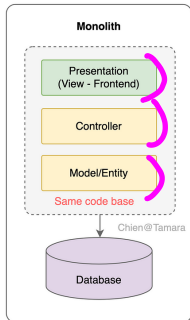
- There are several types of system architectures that are *used* in systems development.
- They include monolithic, client-server, peer-to-peer, and distributed architectures.
- Each *type* of architecture has its own advantages and disadvantages that *depend* on the specific requirements of the system.



Monolithic System Architecture

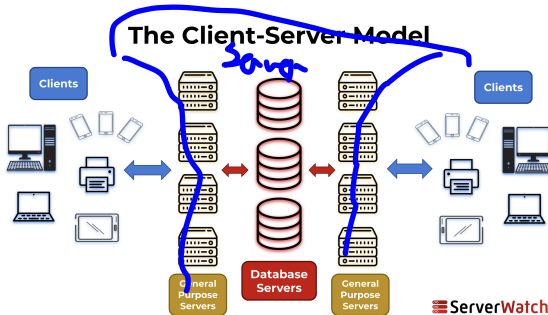
- A **monolithic system architecture** is a **single-tier architecture** that consists of a **single unit** that performs all the functions of the system.
- It is **simple**, **easy** to develop, and **maintain**, but it is **not scalable** and **flexible**. It is typically used for **small systems** that do not require high **performance** or **reliability**.

Java
src
gui
data



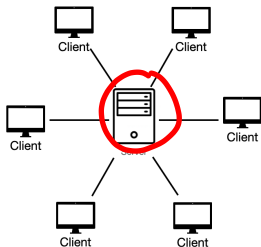
Client-Server System Architecture

- A **client-server system architecture** is a **two-tier architecture** that consists of a **client** and a **server** that **communicate** with each other over a **network**.
- It is **scalable**, **flexible**, and **efficient**, but it is **complex** and **difficult** to **develop** and **maintain**. It is used for **medium** to **large** systems that require **high performance** and **reliability**.

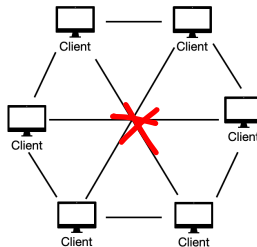


Peer-to-Peer System Architecture

- A **peer-to-peer system architecture** is a **two-tier architecture** that consists of a **network** of **peers** that **communicate** with each other directly.
- It is **scalable**, **flexible**, and **efficient**, but it is **complex** and **difficult** to **develop** and **maintain**. It is used for **medium** to **large** systems that require **high performance** and **reliability**.



Client Server Architecture

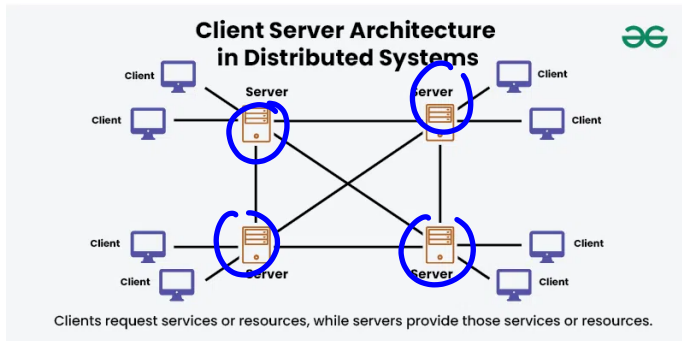


P2P Architecture



Distributed System Architecture

- A **distributed system architecture** is a **multi-tier architecture** that consists of a **network of nodes** that **communicate** with each other over a **network**.
- It is **scalable**, **flexible**, and **efficient**, but it is **complex** and **difficult** to **develop** and **maintain**. It is **used** for **large systems** that **require high performance** and **reliability**.



Software Architectures

- A **software architecture** is a high-level structure of a software system that defines its components, interactions, and relationships.
- A **software architecture** is the blueprint of a software system that guides its development and implementation.
- A **software architecture** is the foundation of a software system that ensures it meets the needs of its users.

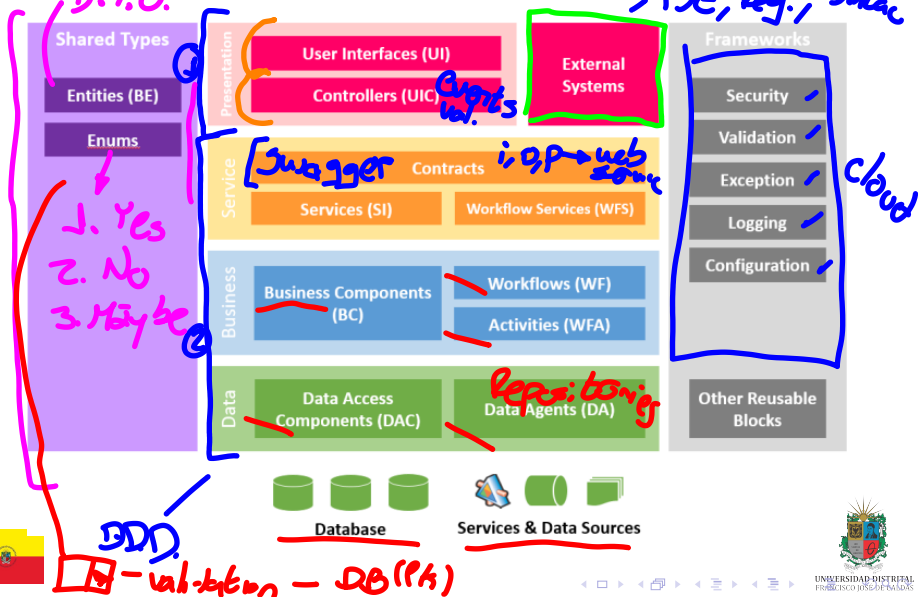
soln of
problem



Layered Architecture Pattern → web - 2009

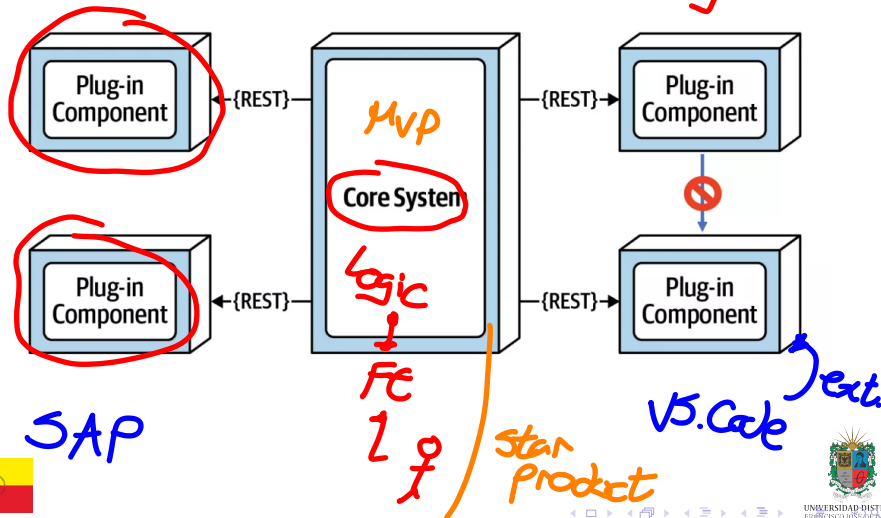
D.T.O.

- PSE, Reg., Data



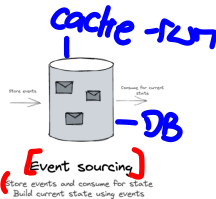
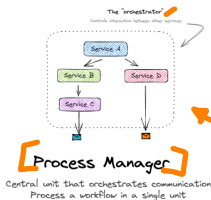
Microkernel Architecture Pattern

Distributed Computing



Event-based Architecture Pattern

To move canvas, hold mouse wheel or spacebar while dragging, or use the hand tool



Inside event-driven architectures

What patterns will you come across when building event-driven architectures?



Point-to-point messaging (pull)

Messages put onto queue by sender
Receiver consumes messages from queue



Change data capture

Listening to events from changes in data
Consume events directly from DB

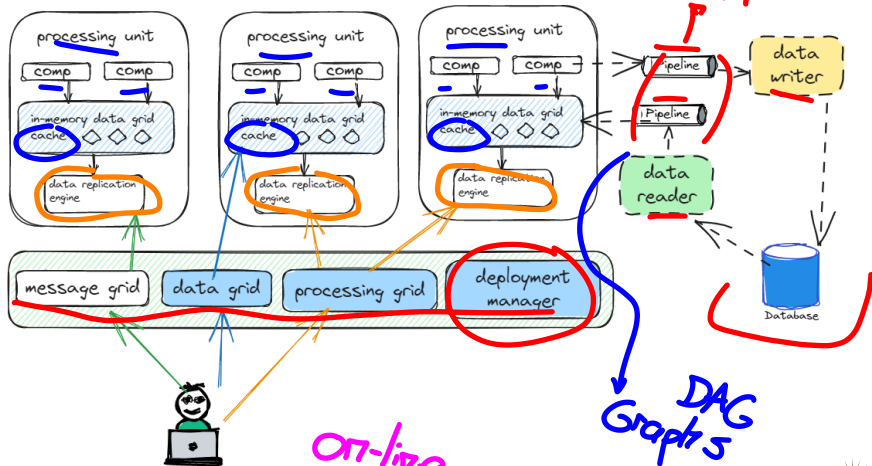


Publish messages/events to many subscribers
Each subscriber gets copy of event to process



Space-based Architecture Pattern

Partial → *shared resources*



on-line video games



Microservices Architecture Pattern



Microservices Architecture on Google Cloud

#GCPsketchnote @PVERGADIA THECLOUDGIRL.DEV 09.12.2021

What are the benefits of microservices?

PROS

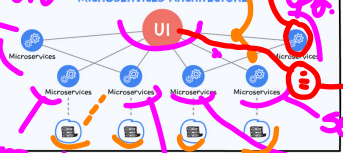


What's the difference between monolith & microservices architecture?

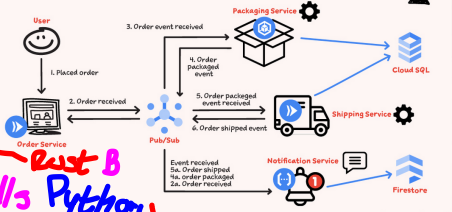
MONOLITHIC ARCHITECTURE



MICROSERVICES ARCHITECTURE

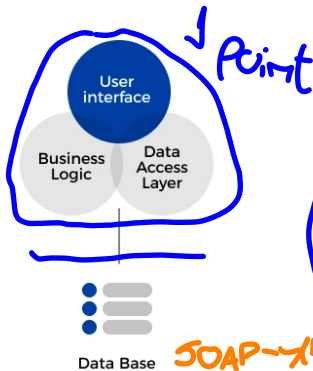


EXAMPLE OF MICROSERVICES ARCHITECTURE IN GOOGLE CLOUD

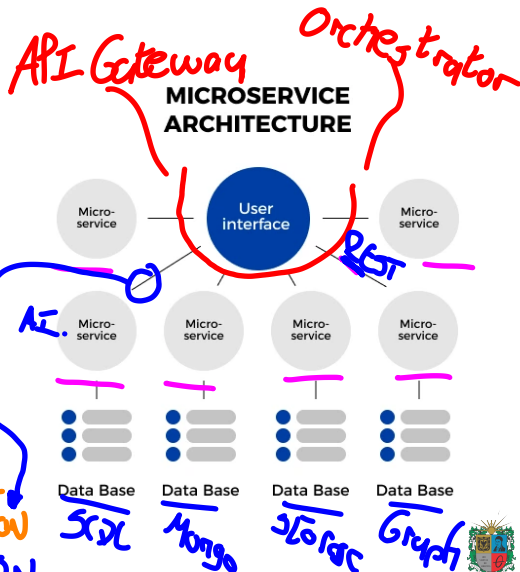


Monolithic vs. Microservices Architectures

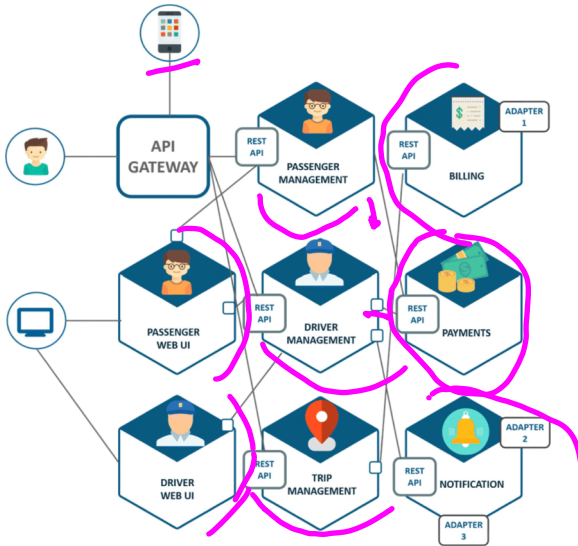
MONOLITHIC ARCHITECTURE



MICROSERVICE ARCHITECTURE



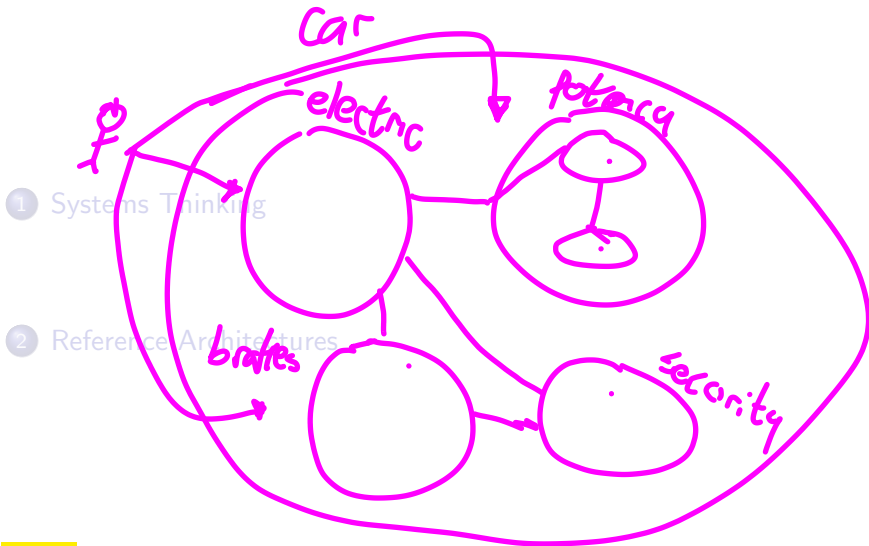
Case Study: Uber Microservices



Source: Kappagantula 2018



Outline



Thanks!

Questions?



Repo: www.github.com/EngAndres/ud-public/tree/main/courses/software_engineering_seminar

